Introduction and Objectives

Since the discovery of diamonds shortly after independence in 1966, diamond revenue has played a central role in Botswana's economic development. It helped finance public infrastructure, education, and healthcare while supporting the country's transition from one of the world's poorest nations to an upper-middle-income economy. This progress, however, created a significant dependence on a single export industry. Although diamonds remain economically important, weaker global demand, competition from lab-grown diamonds, and accumulated inventories have placed increasing pressure on the natural-diamond market and exposed Botswana's limited export diversification (World Bank, 2025; International Monetary Fund [IMF], 2025).

The effects are visible in Botswana's recent economic performance. Real gross domestic product (GDP) contracted by 2.8% in 2024 and by a provisional 0.7% in 2025. In the fourth quarter of 2025, GDP was 5.4% lower than in the same quarter of the previous year, largely because mining and quarrying declined by 47.0%. Over the same period, agriculture grew by 4.2%, manufacturing by 6.5%, and the broader non-mining economy by 2.6% (Statistics Botswana, 2026). These differences suggest that Botswana's diversification problem is not simply the absence of economic activity outside mining. The more relevant question is whether any of the country's non-diamond industries have sufficient scale, stability, and growth potential to become more important sources of export revenue. The urgency of this question was reinforced in March 2026, when S&P Global Ratings lowered Botswana's sovereign credit rating from BBB to BBB− and identified continued weakness in the diamond sector as a major source of pressure on economic growth and public finances (S&P Global Ratings, 2026).

As a citizen of Botswana, my motivation for this project is not abstract. I have watched the diamond industry I grew up associating with national prosperity go from an unquestioned certainty to a genuine source of economic anxiety within the span of a few years. That shift raises a question that matters more to Botswana's government than to any diamond retailer: it is clear that diamonds can no longer carry the economy indefinitely, so what other sectors should the economy be looking? This project is an attempt to answer that question with data rather than intuition.

This project addresses two related questions. First, which non-diamond product categories show the strongest potential for future export growth based on Botswana's historical trade performance? Secondly, how reliably can predictive models identify those opportunities when evaluated using later trade data that was not available during model training? The analysis therefore considers both the size of the predicted opportunity and the reliability of the prediction. The purpose is not to recommend that Botswana abandon diamonds, but to identify which alternative export categories provide the strongest evidence for further investigation and investment.

Data

All data for this project comes directly from UN Comtrade, the United Nations' official international trade database, and was collected using Python through their public API. The dataset contains Botswana's total annual export and import values for every year from 2000 through 2023 in US dollars and is organized by standardized product categories and codes used globally to classify traded goods. After separating diamonds from its broader precious goods category (Category 71), the final dataset returns 98 product categories observed across 24 years, producing 2352 product-year observations.

The dataset identifies the product category, year, annual export value, and annual import value for each observation. Additional variables were constructed in Python to represent each product's historical performance and position within Botswana's trade economy. These include export value lag variables from every five-year period for prediction, annual export growth, a five-year growth trend, a five-year volatility measure, and three-year & five-year averages. The dataset also includes each category's share of Botswana's annual exports and imports, its trade balance, its trade balance relative to total trade, and an indicator to identify whether or not the category exported more than it imported. Together, these variables allow the models to consider a product's existing scale, recent trend, historical stability, and relative importance to Botswana's trade.

Several dollar amount variables were transformed prior to modeling because export values differed substantially across product categories and were strongly right skewed. A small number of categories recorded only a few dollars in certain years, while diamond exports reached several billion dollars. Natural-log transformations were therefore applied to the relevant export, import, lag, and rolling-average variables to reduce the influence of outliers and to allow for a more accurate comparison of products of varying scales. The target variable, next year's product category export value, was also log transformed.

Two data-integrity issues were identified in the data collection phase. First, product category codes below Chapter 10 (for example, "01" for live animals) silently lost their leading zero when the exported CSV was reloaded without explicitly specifying a string data type, a reproducibility risk that had to be corrected at every reload. Second, of the 2,328 product-year observations, 20 had no recorded export value at all; rather than assume these represented a data error, they were treated as instances where UN Comtrade recorded no reported trade activity for

that product in that year, consistent with how the derived lag and growth features were already constructed and were explicitly flagged with an indicator variable rather than silently imputed.

The precious goods category also required additional preprocessing. The standard category covering precious stones and metals combines diamonds with several other products. Because diamonds dominate Botswana's exports but are not a diversification opportunity for the purposes of this project, this category was separated into two groups: diamonds and all other precious goods. Both groups were retained during model development so that the models could learn from Botswana's complete export history. Diamonds were excluded only when producing the final recommendations. The small miscellaneous category for commodities not otherwise classified was also excluded from those recommendations because it does not represent a clearly identifiable industry. All other precious goods remained eligible for consideration.

Because a lot of the predictors, including the five annual export lags and five-year trend & volatility measures, required a complete five-year history, observations from 2000 through 2004 could not be used directly for modeling. Instead, these years were retained as historical context for constructing the predictor values available from 2005 onward. The eligible observations were then split chronologically with an 80/20 split. Using an earlier period for training and a later period for validation preserved the order of the data and evaluated how well the models predicted future export outcomes.

One limitation of the dataset is that it ends in 2023 as opposed to 2025, the most recent calendar year, as this was the most recent year for which Botswana's trade data was available from UN Comtrade. I tried to search for these values from other official sources, however I realized that I was not going to find this level of detail regarding Botswana's imports and exports, so for consistency and accuracy of the data, I chose to leave the data as is. Consequently,

the 2023 observations served as the inputs for genuinely forward-looking predictions of 2024 export values. After the models had been finalized and the predictions generated, I was able to obtain official 2024 and 2025 trade figures from Statistics Botswana, however the data was not anywhere near as detailed as that from Un Comtrade. The directly comparable values were then used as an independent assessment of the SAS developed forecasts against subsequently realized outcomes. Because these figures were not used during model training, validation, or selection, this comparison provides an additional out-of-sample test of the models' practical performance.

Model Development

The training dataset was imported into SAS, where a partition variable built in Python split the 1,764 eligible development rows into 1,372 training observations (2005–2018) and 392 validation observations (2019–2022), using a 80/20 split. SAS Viya's default partitioning assigns rows at random, which would let a model train on years that might fall after some of its validation years, a form of leakage for time-dependent trade data. By supplying a pre-built chronological partition and disabling automatic partitioning, this potential issue was completely avoided. A Transformation node log-transformed eleven nonnegative dollar-denominated predictors (current exports/imports, three- and five-year averages, the five annual lags, and total national exports/imports); trade balance and percentage, growth, trend, volatility, count, and binary predictors were left on their original scales. An Imputation node then addressed the 17 current-year export values that remained missing by giving them the number 0, as any other imputation technique would have given incorrect results.

Eight models were trained on this data and compared by ASE: full linear regression, stepwise and LASSO regressions, a decision tree, random forest, gradient boosting, neural network, and SVM. Comparing regression, tree, and margin-based models was the only way to

see which one generalized to unseen years rather than just fitting the training period. The full regression model, using all 22 predictors, showed why a more disciplined approach was needed: several lag/average export variables carried variance inflation factors into the hundreds (168.9 for the two-year log export lag), since a product's current exports, lags, and rolling averages are naturally correlated. Its most significant predictors were annual import value, an indicator for no recorded exports, the two-year export lag, and imputed current exports (all $p < .001$); validation ASE was 3.1779. Stepwise regression improved this to 2.8176, and LASSO, which shrinks weaker coefficients via an L1 penalty rather than dropping them outright, reached 2.7722 (train ASE 1.6716), the strongest fully-interpretable model in the comparison.

Tree-based methods performed worse. The decision tree (21 leaves, cost-complexity pruned, max depth 10) had the weakest validation ASE of all eight models (3.4728); The forest model reached 3.2714 and gradient boosting 3.4144. These results suggest that once export history is log-transformed, the strongest signal, a product's own export level and trend, behaves fairly linearly, which splitting rules exploit less efficiently than a coefficient does. The neural network (two hidden layers of 15 neurons, L2 = 0.001) reached a validation ASE of 2.9201 and R-Square of 0.8030 (train ASE 1.5592, train R^2 0.8469); an earlier, more heavily regularized version had collapsed toward predicting the sample mean for every product, and reducing the L2 penalty restored genuine learning. The SVM, with a linear kernel, penalty C = 0.1, and early stopping, produced the lowest validation ASE of all eight models: 2.7583 (root ASE 1.6608) and R-Square of 0.8139.

Model Comparison and Selection

As you will see in Table 1, SVM was the statistical champion on validation ASE; LASSO trailed by only 0.0139 points (about 0.5%), making it nearly equivalent, but contained more transparent predictors and is easier to interpret. For the purposes of the project, SVM was selected for final scoring and LASSO was used as an interpretive check on whether the SVM's forecast for a category is corroborated by an independent model. The other models did not perform as well in comparison: linear regression's 3.1779 confirms that multicollinearity hurt the prediction quality of the model in relation to LASSO's regularized version. The decision tree was the weakest of any model, and the neural network placed fourth, ahead of every tree-based method but behind the three linear approaches. This validation-based selection was the correct choice at the time it was made, using only information available before 2024; whether it held up against real subsequent trade data is examined below.

Scoring and Results

With SVM selected as the winning model, the scoring dataset was imported. This dataset contained 98 product categories with complete 2023 data, including the separated 7102 diamond and 71_OTHER precious-goods codes and was scored with the target being predicted_2024_usd. Diamonds (HS 7102) and Category 99 were excluded from the recommendations because diamonds are being used as the concentration problem that this project examines as opposed to being used as a diversification target, and Category 99 is essentially a miscellaneous, unidentifiable grouping which therefore results in no specific “product sector” that we can recommend is used as a diversifier for Botswana’s economy. All other precious goods traded under the broader category 71 were kept in the recommendation because they provide legitimate

diversification options. The remaining categories were ranked by predicted growth, and required a value of at least $100,000 in 2023 exports so tiny categories could not dominate on a low base.

As you will see in Table 2, Category 81 (base metals, cermets and related articles) ranks first at +140.5%, but is actually a genuine red flag rather than a legitimate recommendation. Its five-year log volatility of 2.889 is far higher than any other eligible category, driven by a single-year spike to nearly $3 million in 2022 followed by a roughly 90% collapse in 2023. A model trained partly on that spike projects reversion toward it, but one anomalous year is not the repeatable momentum that a low-volatility category like cereals or sugars and confectionery can offer, so Category 81 is excluded despite ranking first, leaving the remaining eight rows of Table 2 as the retained candidates

Because percentage growth favors small categories that can double from a low base, Botswana's largest existing non-diamond exporters were also checked against LASSO's independent forecast. Mineral fuels and oils, already a $36.7 million category in 2023, combined that scale with a 33.6% SVM forecast and a 20.8% LASSO forecast in the same direction, one of the clearest cases of cross-model agreement in the dataset. Machinery ($40.6M), vehicles ($27.4M), and pharmaceuticals ($11.9M) showed similar agreement. LASSO forecast declines where SVM forecast growth for electrical machinery ($181.4M), inorganic chemicals ($56.5M), live animals ($45.7M), and 71_OTHER ($23.8M), so confidence in those four should be lower than for the categories where both models agreed.

External Validation Checks

Because SAS Viya's SVM isn't directly built for multi-year forecasting, an independent Python model was used to check both the SAS output and how far its predictive value extends. A

standardized LinearSVR model, matched to the SAS SVM's specification and using the same chronological validation split, produced 2024 predictions that correlated at r = 0.9995 with the saved SAS SVM scores across all 98 products, confirming the SAS results are reproducible in direction and magnitude. The same framework was extended to predict directly at horizons of one through five years, preserving the same chronological validation at every horizon. Validation R-Square declined from 0.8064 one year out to roughly 0.67–0.69 by four and five years out, while validation ASE rose from 2.8703 to 4.7369 (Table 3), reflecting the smaller and more distant samples available at longer horizons rather than a genuine reversal in accuracy.

After the SAS models were finalized, official 2024 and 2025 trade figures were obtained directly from Statistics Botswana's International Merchandise Trade Statistics report and converted to US dollars using Bank of Botswana exchange rates, providing a genuine, previously unseen test of both the SVM and LASSO forecasts. The results reversed the internal validation ranking. Table 4 compares each model's 2024 forecast against the real outcome across total exports, diamond exports, and non-diamond exports. LASSO's total-export forecast missed by only 2.97%, in the correct direction, while SVM's missed by 37.69% in the wrong direction. Across all three components, LASSO's mean absolute percentage error was 26.4% with two of three directions correct, compared to SVM's 45.3% with zero of three directions correct. Despite having the lower validation ASE inside SAS, the SVM was measurably less reliable once tested against real subsequent trade data, while LASSO, the close runner-up in Table 1, held up better. This does not overturn the validation-based selection, since validation ASE was the only criterion available at the time a model had to be chosen without future data, but it is a meaningful caveat: the champion model by internal validation was not the champion by external, real-world accuracy.

Both models also underestimated non-diamond export growth substantially in 2024 and 2025. LASSO forecast a 38.7% decline in non-diamond exports for 2024; the actual figure grew 22.3%. A 2025 extension of the same model repeated the pattern, forecasting a further 38.5% decline against actual growth of 28.9%. This is the opposite of what this project's diversification argument would need: rather than overstating non-diamond potential, the models understated it, which if anything strengthens the case that non-diamond sectors are worth continued attention, even though it means the specific growth figures in Table 2 should be read as conservative rather than precise.

Recommendations and Limitations

These results support a specific recommendation. Two groups of non-diamond related categories stand out. The first is a high-confidence group, built on existing scale and cross-model agreement rather than raw growth: mineral fuels and oils ($36.7M, SVM +33.6%, LASSO +20.8%), machinery ($40.6M), vehicles ($27.4M), and pharmaceuticals ($11.9M), all cases where SVM and LASSO independently agreed on the direction of growth. The second is a high-growth group drawn from Table 2, smaller in current scale but forecast to grow fastest with comparatively low volatility: miscellaneous manufactured articles, cereals, sugars and confectionery, ceramic products, beverages and spirits, wood pulp, and printed books. Categories where SVM and LASSO disagreed, namely electrical machinery, inorganic chemicals, live animals, and non-diamond precious goods, are not recommended here; they may still be worth investigating with more extensive data, but not on this analysis's evidence alone.

It is important to note that none of these recommended categories identifies a single replacement for diamonds. Diamonds have accounted for roughly 63% to 91% of Botswana's exports for many years and no non-diamond category in this dataset even comes close to that

scale. The recommendation above is a portfolio of candidates for further investigation, not a substitute.

In thinking about what could improve this analysis, four things stand out. First, and most importantly, the model that wins on internal validation is not guaranteed to be the model that performs best going forward: external validation against real 2024–2025 outcomes showed LASSO clearly outperforming the SAS validation champion, SVM, an important caution against over-trusting a single validation metric. Second, both models substantially underestimated actual non-diamond export growth in 2024 and 2025, so the specific growth percentages in this paper should be read as conservative rather than precise, even though the general direction, that non-diamond sectors have real growth potential, held up. Third, UN Comtrade data only goes up to 2023, so this project cannot generate a genuinely forward-looking prediction beyond what's already been checked here. Finally, these findings remain predictive rather than causal: a strong forecast means a category's history resembles that of categories that grew, not that a specific policy would cause future growth, so this is a starting point for further investigation rather than a basis for direct investment.

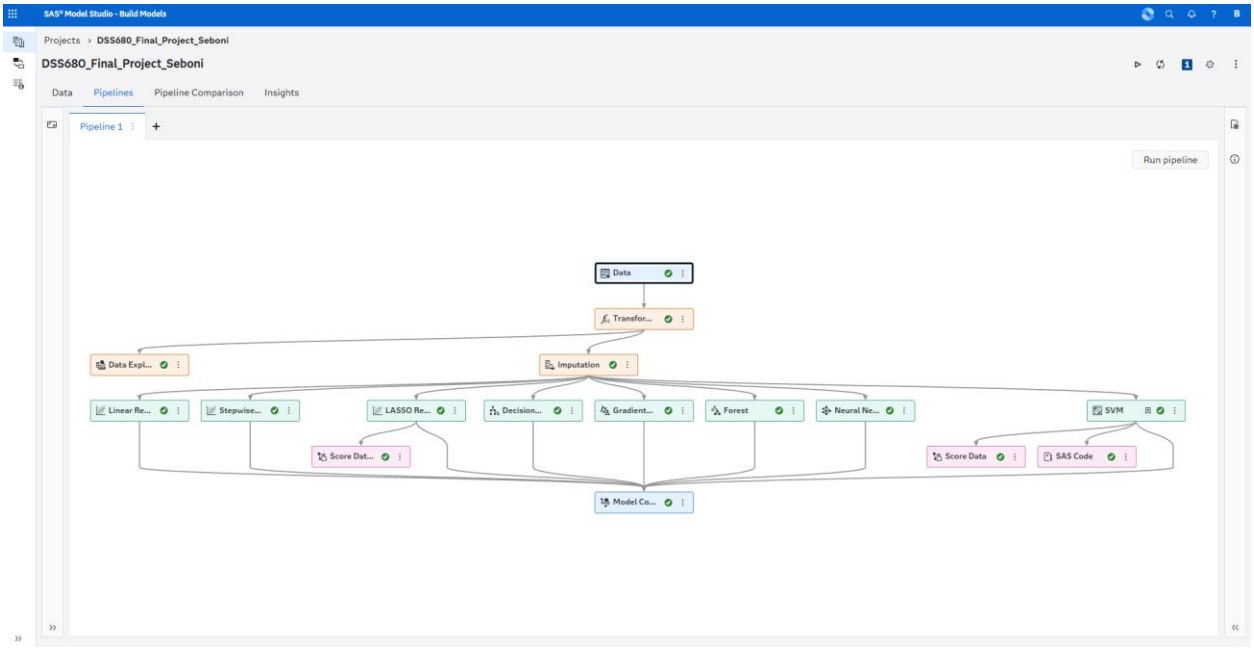
Conclusion

This project asked which of Botswana's non-diamond export categories show the strongest evidence of contributing more meaningfully to the country's export portfolio, and how much confidence that evidence deserves. Eight models, evaluated on a chronological, leakage-resistant design, point to the same two groups identified above: mineral fuels, machinery, vehicles, and pharmaceuticals, all backed by scale and cross-model agreement, and a smaller-scale, high-growth group led by miscellaneous manufactured articles, cereals, and sugars and confectionery. This analysis does not offer Botswana a direct a replacement for diamonds; no

category found in this data will realistically come close to that scale, especially not within the near future. The analysis provides a data-driven starting point for the more challenging conversations that Botswana's government is hopefully already having about where to redirect investment towards so that the country never again finds themselves relying primarily on this one product for economic prosperity.

This paper was structured and formatted with the assistance of Claude.

Final SAS Pipeline



Appendices

Table 1 – Model Comparison Results

Model Comparison

Champion	Name	Algorithm Name	Average Squared Error	Data Role	Number of Observations	Root Average Squared ...	Root Mean Absolute Er...	Root Mean Squared Lo...
*	SVM	SVM	2.7583	VALIDATE	392	1.6608	0.9951	0.3236
	Linear Regression	Linear Regression	3.1779	VALIDATE	392	1.7827	1.0267	0.3508
	Stepwise Regression	Linear Regression	2.8176	VALIDATE	392	1.6786	1.0062	0.3380
	LASSO Regression	Linear Regression	2.7722	VALIDATE	392	1.6650	1.0067	0.3149
	Decision Tree	Decision Tree	3.4728	VALIDATE	392	1.8635	1.0642	0.3432
	Gradient Boosting	Gradient Boosting	3.4144	VALIDATE	392	1.8478	1.0531	0.3605
	Forest	Forest	3.2714	VALIDATE	392	1.8087	1.0333	0.3582
	Neural Network	Neural Network	2.9201	VALIDATE	392	1.7088	1.0127	0.3378

Table 2 – Diversification grwoth

Code	Product Category	2023 Exports	SVM 2024 Forecast	Predicted Growth
81	Metals, n.e.c., cermets and articles thereof*	$309,286	$743,888	+140.5%
96	Miscellaneous manufactured articles	$107,116	$196,391	+83.3%
10	Cereals	$839,478	$1,527,147	+81.9%
17	Sugars and sugar confectionery	$332,216	$561,834	+69.1%
69	Ceramic products	$125,255	$173,291	+38.4%
27	Mineral fuels and mineral oils	$36,738,787	$49,090,809	+33.6%
22	Beverages, spirits and vinegar	$1,956,453	$2,542,725	+30.0%
47	Wood pulp and recovered paper	$879,960	$1,127,243	+28.1%
49	Printed books, newspapers, pictures	$110,557	$139,150	+25.9%

Table 3 – Multi-horizon Validation Results

Horizon	Target Year	Training Rows	Validation Rows	Validation ASE	Validation R^2
1 year	2024	1,372	392	2.8703	0.8064
2 years	2025	1,274	392	3.7393	0.7477
3 years	2026	1,176	392	4.5495	0.6931
4 years	2027	1,176	294	4.3491	0.6956
5 years	2028	1,078	294	4.7369	0.6685

Table 4 – External Validation results

Model	Export Component	2023 Baseline	Predicted 2024	Actual 2024	Abs. % Error	Direction Correct
LASSO	Total exports	$5,502.7M	$4,803.4M	$4,665.0M	2.97%	Yes
SVM	Total exports	$5,502.7M	$6,423.3M	$4,665.0M	37.69%	No
LASSO	Diamond exports (HS 7102)	$4,333.6M	$4,086.9M	$3,234.9M	26.34%	Yes
SVM	Diamond exports (HS 7102)	$4,333.6M	$5,428.3M	$3,234.9M	67.80%	No
LASSO	Non-diamond exports	$1,169.2M	$716.5M	$1,430.1M	49.90%	No
SVM	Non-diamond exports	$1,169.2M	$995.0M	$1,430.1M	30.42%	No

SVM Results

Output

The SAS System

The SVMACHINEProcedure

Model Information	
Task Type	ESP_REG
Optimization Technique	InteriorPoint
Scale	NO
Kernel Function	Linear
Penalty Method	C
Penalty Parameter	0.1
Insensitive Loss Parameter	0.01
Maximum Iterations	25
Tolerance	1e-06

Number of Observations Read	1764
Number of Observations Used	1764
Number of Observations Used for Training	1372
Number of Observations Used for Validation	392

Training Results	
Inner Product of Weights	1.44616802
Bias	10.1967636
Total Slack (Constraint Violations)	1040.42511
Norm of Longest Vector	4726410209
Mean Squared Error	1.61560208
Number of Effects	22
Columns in Data Matrix	25

Predicted Target Variable
Variable
P_next_year_log_export_value

Output CAS Tables			
CAS Library	Name	Number of Rows	Number of Columns
Analytics_Project_d178faeea136-4640-a932-06cffc1e71cd	_90D26YKBMOK7NRMK41P3SW06Z_AST	1	2

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
next_year_log_export_value	TRAIN	0	0
next_year_log_export_value	VALIDATE	1	1

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,372	1.6156	1,372	1.2711
392	2.7583	392	1.6608

Mean Absolute Error	Root Mean Absolute Error	Mean Squared Logarithmic Error	Root Mean Squared Logarithmic Error
0.7683	0.8765	0.0329	0.1815
0.9903	0.9951	0.1047	0.3236

R-Square
0.8414
0.8139

Training Results

Statistic	Description	Value
WW	Inner Product of Weights	1.4462
Beta	Bias	10.1968
TotalSlack	Total Slack (Constraint Violations)	1,040.4251
LongVector	Norm of Longest Vector	4.72641021E9
MSE	Mean Squared Error	1.6156
nEffects	Number of Effects	22
nLevels	Columns in Data Matrix	25

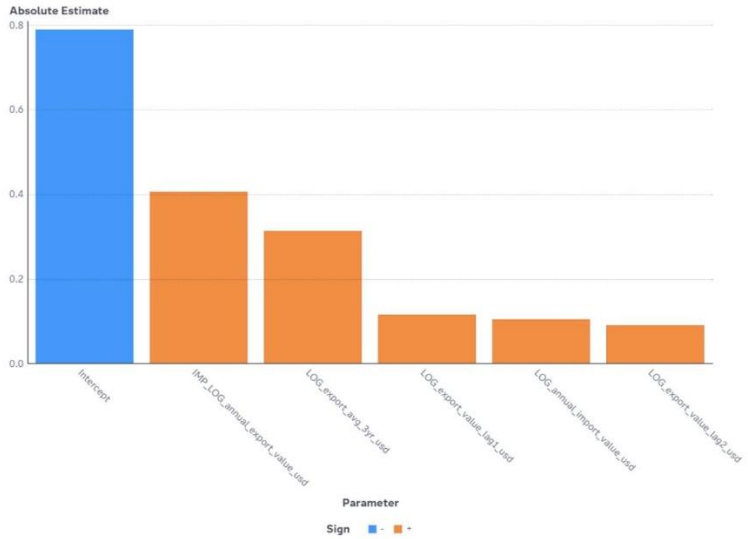
LASSO Outputs

Selection Summary

Step	Effect Entered	Number of Effects	Validation ASE
0	Intercept	1	15.1693
1	LOG_export_avg_3yr_usd	2	3.2981
2	IMP_LOG_annual_export_value_usd	3	2.8308
3	LOG_export_value_lag1_usd	4	2.8056
4	LOG_export_value_lag2_usd	5	2.8125
5	LOG_annual_import_value_usd	6	2.7722
6	no_recorded_exports_1	7	2.8434
7	LOG_total_exports_all_products_u	8	2.9062
8	is_net_exporter_1	9	2.9122

Optimal Validation ASE
0
0
0
0
0
1
0
0
0

Parameter Estimates



This plot displays the absolute value of the estimate for each parameter in the model when using LASSO selection. The bar that represents the parameter is colored by the sign of the estimate. Bars that are colored as positive (+) correspond to a positive parameter estimate, which indicates an increase in the predicted target value as the parameter value increases. Bars that are colored as negative (-) correspond to a negative parameter estimate, which indicates a decrease in the predicted target value as the parameter value increases. The parameter with the largest effect is Intercept with an estimate of -0.79.

Parameter Estimates

Effect	Parameter	Sign	Estimate
Intercept	Intercept	-	-0.7896
IMP_LOG_annual_export_value_usd	IMP_LOG_annual_export_value_usd	+	0.4063
LOG_export_avg_3yr_usd	LOG_export_avg_3yr_usd	+	0.3135
LOG_export_value_lag1_usd	LOG_export_value_lag1_usd	+	0.1155
LOG_annual_import_value_usd	LOG_annual_import_value_usd	+	0.1047
LOG_export_value_lag2_usd	LOG_export_value_lag2_usd	+	0.0905

Absolute Estimate	Degrees of Freedom
0.7896	1
0.4063	1
0.3135	1
0.1155	1
0.1047	1
0.0905	1

Regression Fit Statistics

Statistic	Description	Value
RMSE	Root MSE	1.2958
RSQUARE	R-Square	0.8359
ADJRSQ	Adj R-Sq	0.8353
AIC	AIC	2,090.9365
AICC	AICC	2,091.0186
SBC	SBC	748.2807
TRAIN_ASE	ASE (Train)	1.6716
VAL_ASE	ASE (Validate)	2.7722

Linear Regression Output

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
next_year_log_export_value	TRAIN	0	0
next_year_log_export_value	VALIDATE	1	1

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,372	1.5327	1,372	1.2380
392	3.1779	392	1.7827

Mean Absolute Error	Root Mean Absolute Error	Mean Squared Logarithmic Error	Root Mean Squared Logarithmic Error
0.7806	0.8835	0.0307	0.1754
1.0540	1.0267	0.1231	0.3508

R-Square
0.8495
0.7856

Parameter Estimates

Effect	Parameter	t Value	Sign
LOG_annual_import_value_usd	LOG_annual_import_value_usd	6.7406	+
no_recorded_exports	no_recorded_exports 0	4.5552	-
LOG_export_value_lag2_usd	LOG_export_value_lag2_usd	4.5033	+
IMP_LOG_annual_export_value_usd	IMP_LOG_annual_export_value_usd	4.2684	+
Intercept	Intercept	3.5855	+
normalized_trade_balance	normalized_trade_balance	3.3997	+
recorded_export_years_5yr	recorded_export_years_5yr	3.3528	-
export_log_growth_1yr	export_log_growth_1yr	2.6513	-
export_log_trend_5yr	export_log_trend_5yr	1.9230	+
LOG_export_avg_3yr_usd	LOG_export_avg_3yr_usd	1.6204	-
LOG_export_avg_5yr_usd	LOG_export_avg_5yr_usd	1.5491	+
LOG_total_exports_all_products_u	LOG_total_exports_all_products_u	1.4081	-
is_diamond_category	is_diamond_category 0	1.3565	+
LOG_total_imports_all_products_u	LOG_total_imports_all_products_u	1.2915	-
export_share_pct	export_share_pct	1.2720	+
is_net_exporter	is_net_exporter 0	1.0330	+
import_share_pct	import_share_pct	0.9630	-
export_log_volatility	export_log_volatility	0.8932	-

Effect	Parameter	t Value	Sign
_5yr	_5yr		
product_trade_balance_usd	product_trade_balance_usd	0.6489	-
LOG_export_value_lag3_usd	LOG_export_value_lag3_usd	0.3708	-
LOG_export_value_lag5_usd	LOG_export_value_lag5_usd	0.2616	+

Estimate	Absolute Estimate	Standard Error	Pr > |t|
0.2936	0.2936	0.0436	0.0000
-2.8924	2.8924	0.6350	0.0000
0.1867	0.1867	0.0415	0.0000
0.5131	0.5131	0.1202	0.0000
14.9484	14.9484	4.1691	0.0003
0.9652	0.9652	0.2839	0.0007
-1.0664	1.0664	0.3180	0.0008
-0.1432	0.1432	0.0540	0.0081
0.4095	0.4095	0.2130	0.0547
-0.1807	0.1807	0.1115	0.1054
0.2485	0.2485	0.1604	0.1216
-0.3090	0.3090	0.2194	0.1593
3.1911	3.1911	2.3524	0.1752
-0.1829	0.1829	0.1416	0.1968
0.0569	0.0569	0.0447	0.2036
0.3084	0.3084	0.2985	0.3018
-0.0260	0.0260	0.0270	0.3357
-0.1101	0.1101	0.1233	0.3719
0.0000	0.0000	0.0000	0.5165
-0.0135	0.0135	0.0363	0.7109
0.0064	0.0064	0.0246	0.7937

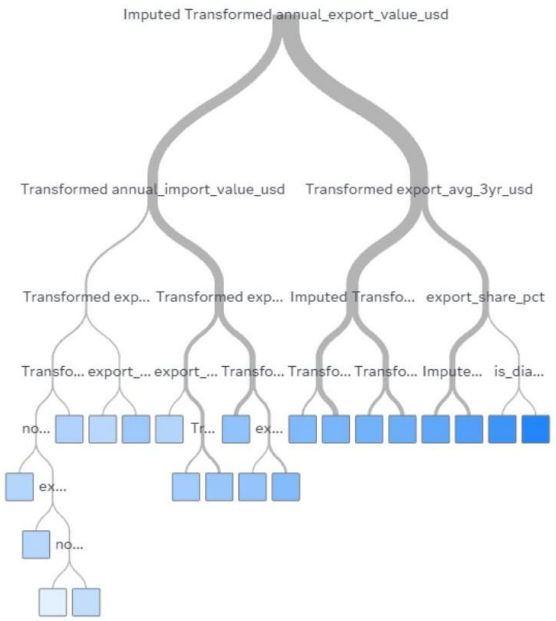
Variance Inflation	Degrees of Freedom
8.6258	1
1.8039	1
15.4157	1
130.1591	1
0	1
18.5282	1
2.0311	1
5.9665	1
9.0378	1
87.5387	1
168.8968	1
2.5612	1
49.2635	1
2.5529	1
108.8918	1
7.8709	1
6.0954	1
9.5440	1
18.1473	1
11.7198	1
5.4716	1

Regression Fit Statistics

Statistic	Description	Value
RMSE	Root MSE	1.2476
RSQUARE	R-Square	0.8495
ADJRSQ	Adj R-Sq	0.8473
AIC	AIC	2,001.8890
AICC	AICC	2,002.6392
SBC	SBC	737.5935
TRAIN_ASE	ASE (Train)	1.5327
VAL_ASE	ASE (Validate)	3.1779

Decision Tree Output

Tree Diagram



Fit Statistics

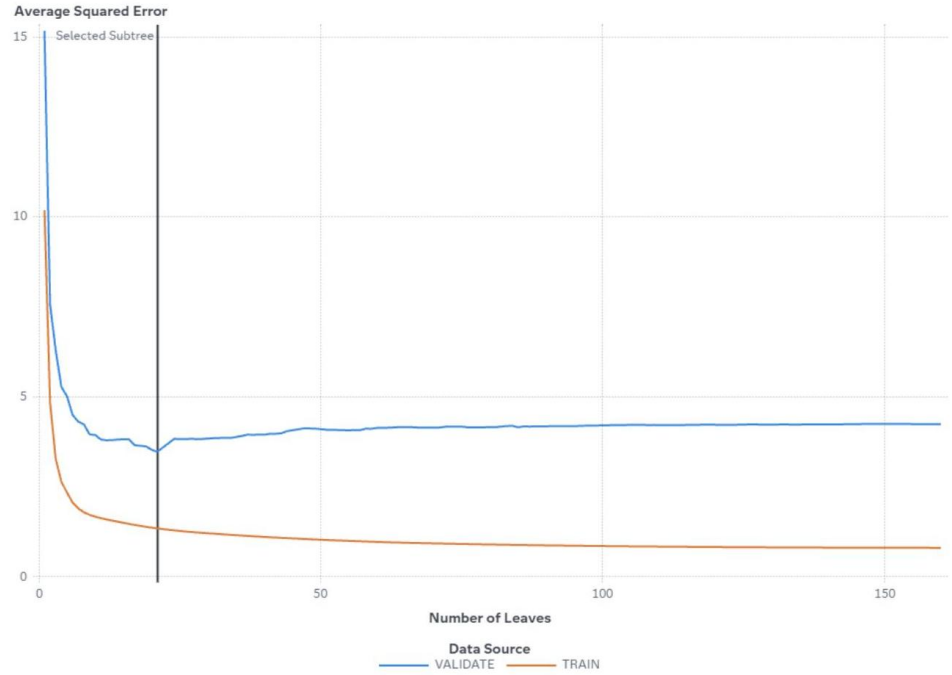
Target Name	Data Role	Partition Indicator	Formatted Partition
next_year_log_export_value	TRAIN	0	0
next_year_log_export_value	VALIDATE	1	1

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,372	1.3463	1,372	1.1603
392	3.4728	392	1.8635

Mean Absolute Error	Root Mean Absolute Error	Mean Squared Logarithmic Error	Root Mean Squared Logarithmic Error
0.7831	0.8849	0.0218	0.1476
1.1325	1.0642	0.1178	0.3432

R-Square
0.8678
0.7657

Pruning Error Plot



This plot shows how the average squared error changes for subtrees, which are created by pruning the full decision tree to various numbers of leaves. The training error decreases as the number of leaves increases, so the VALIDATE partition can be used to prune the tree to prevent overfitting. For this decision tree model, the selected subtree based on the pruning options has 21 leaves with an average squared error of 3.473 for the VALIDATE partition.

Variable Importance

Variable Name	Training Importance	Training Relative Importance	Validation Relative Importance
IMP_LOG_annual_export_value_usd	7,717.8142	1	1
LOG_export_avg_3yr_usd	2,663.1634	0.3451	0.2343
LOG_annual_import_value_usd	867.8913	0.1125	0.1286
export_share_pct	530.1416	0.0687	0.0754
LOG_export_value_lag2_usd	54.8272	0.0071	0.0159
export_log_volatility_5yr	56.0847	0.0073	0.0129
LOG_export_avg_5yr_usd	31.0049	0.0040	0.0122
LOG_export_value_lag3_usd	37.7572	0.0049	0.0038
is_diamond_category	51.2479	0.0066	0.0031
normalized_trade_balance	74.8520	0.0097	0.0005
export_log_trend_5yr	40.6465	0.0053	0.0001

Count	Validation Importance
3	3,083.8583
5	722.4571
1	396.5738
3	232.5111
1	49.0282

Stepwise Regression Output

Output

The SAS System

The REGSELECT Procedure

Model Information	
Data Source	_INPUT_B5XHDXE0DM1H5I48PTVYMEK41
Response Variable	next_year_log_export_value
Predicted Response	P_next_year_log_export_value

Selection Information	
Selection Method	Stepwise
Select Criterion	SBC
Choose Criterion	ValidationASE
Stop Criterion	ValidationASE
Effect Hierarchy Enforced	None
Stop Horizon	3

Number of Observations Read	1764
Number of Observations Used	1764
Number of Observations Used for Training	1372
Number of Observations Used for Validation	392
Number of Observations Used for Testing	0

Class Level Information		
Class	Levels	Values
is_diamond_category	2	0 1
is_net_exporter	2	0 1
no_recorded_exports	2	0 1

Dimensions	
Number of Effects	23
Number of Parameters	26

Regression Fit Statistics

Statistic	Description	Value
RMSE	Root MSE	1.2832
RSQUARE	R-Square	0.8389
ADJRSQ	Adj R-Sq	0.8384
AIC	AIC	2,063.2088
AICC	AICC	2,063.2703
SBC	SBC	715.3289
TRAIN_ASE	ASE (Train)	1.6406
VAL_ASE	ASE (Validate)	2.8176

Selection Summary

Step	Effect Entered	Number of Effects	SBC
0	Intercept	1	3,191.3965
1	LOG_export_avg_3yr_usd	2	935.4933
2	IMP_LOG_annual_export_value_usd	3	782.3773
3	no_recorded_exports	4	747.4693
4	LOG_annual_import_value_usd	5	715.3289
5	normalized_trade_balance	6	700.6046
6	LOG_export_value_lag2_usd	7	695.3700
7	LOG_total_exports_all_products_u	8	686.3481

Optimal SBC	Validation ASE	Optimal Validation ASE
0	15.1693	0
0	3.2981	0
0	2.8308	0
0	2.8668	0
0	2.8176	1
0	2.8252	0
0	2.8717	0
1	2.9190	0

Parameter Estimates

Effect	Parameter	t Value	Sign
IMP_LOG_annual_export_value_usd	IMP_LOG_annual_export_value_usd	13.6502	+
LOG_export_avg_3yr_usd	LOG_export_avg_3yr_usd	10.3443	+
no_recorded_exports	no_recorded_exports 0	6.6229	-
LOG_annual_import_value_usd	LOG_annual_import_value_usd	6.3079	+
Intercept	Intercept	4.1014	+

Estimate	Absolute Estimate	Standard Error	Pr > |t|
0.5128	0.5128	0.0376	0.0000
0.4188	0.4188	0.0405	0.0000
-3.8375	3.8375	0.5794	0.0000
0.1262	0.1262	0.0200	0.0000
2.5789	2.5789	0.6288	0.0000

Variance Inflation	Degrees of Freedom
12.0170	1
10.9052	1
1.4200	1
1.7206	1
0	1

Gradient Boosting Model Output

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
next_year_log_export_value	TRAIN	0	0
next_year_log_export_value	VALIDATE	1	1

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,372	1.0451	1,372	1.0223
392	3.4144	392	1.8478

Mean Absolute Error	Root Mean Absolute Error	Mean Squared Logarithmic Error	Root Mean Squared Logarithmic Error
0.6810	0.8252	0.0233	0.1525
1.1089	1.0531	0.1300	0.3605

R-Square
0.8974
0.7697

Variable Importance

Variable Label	Role	Variable Name	Training Importance
	INPUT	export_share_pct	360.5539
Imputed Transformed annual_export_value_usd	INPUT	IMP_LOG_annual_export_value_usd	272.4783
Transformed export_avg_3yr_usd	INPUT	LOG_export_avg_3yr_usd	246.8134
Transformed export_value_lag1_usd	INPUT	LOG_export_value_lag1_usd	53.4736
Transformed export_avg_5yr_usd	INPUT	LOG_export_avg_5yr_usd	33.1720
Transformed export_value_lag2_usd	INPUT	LOG_export_value_lag2_usd	32.2558
Transformed annual_import_value_usd	INPUT	LOG_annual_import_value_usd	31.4435
	INPUT	product_trade_balance_usd	15.4064
	INPUT	import_share_pct	13.1288
Transformed export_value_lag3_usd	INPUT	LOG_export_value_lag3_usd	12.4062
Transformed export_value_lag5_usd	INPUT	LOG_export_value_lag5_usd	10.1551
	INPUT	export_log_volatility_5yr	10.0659
	INPUT	export_log_trend_5yr	9.7840

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
next_year_log_export_value	TRAIN	0	0
next_year_log_export_value	VALIDATE	1	1

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,372	0.9861	1,372	0.9930
392	3.2714	392	1.8087

Mean Absolute Error	Root Mean Absolute Error	Mean Squared Logarithmic Error	Root Mean Squared Logarithmic Error
0.5995	0.7743	0.0245	0.1566
1.0677	1.0333	0.1283	0.3582

R-Square
0.9032
0.7793

Variable Importance

Variable Label	Role	Variable Name	Training Importance
	INPUT	export_share_pct	1,677.8890
Imputed Transformed annual_export_value_usd	INPUT	IMP_LOG_annual_export_value_usd	1,451.2620
Transformed export_avg_3yr_usd	INPUT	LOG_export_avg_3yr_usd	1,316.0515
Transformed export_avg_5yr_usd	INPUT	LOG_export_avg_5yr_usd	955.1080
Transformed export_value_lag1_usd	INPUT	LOG_export_value_lag1_usd	701.7482
Transformed export_value_lag3_usd	INPUT	LOG_export_value_lag3_usd	432.0875
Transformed export_value_lag2_usd	INPUT	LOG_export_value_lag2_usd	290.8661
Transformed annual_import_value_usd	INPUT	LOG_annual_import_value_usd	158.7205
Transformed export_value_lag4_usd	INPUT	LOG_export_value_lag4_usd	134.0643
	INPUT	product_trade_balance_usd	111.6696
	INPUT	import_share_pct	109.5726
Transformed export_value_lag5_usd	INPUT	LOG_export_value_lag5_usd	94.1847
	INPUT	export_log_trend_5y	62.1206

Neural Network Output

Output

The SAS System

The NNET Procedure

Model Information	
Model	NeuralNet
Number of Observations Used	1372
Number of Observations Read	1372
Target/Response Variable	next_year_log_export_value
Number of Nodes	56
Number of Input Nodes	25
Number of Output Nodes	1
Number of Hidden Nodes	30
Number of Hidden Layers	2
Number of Weight Parameters	615
Number of Bias Parameters	31
Architecture	MLP
Seed for Initial Weight	12345
Optimization Technique	LBFGS
Number of Neural Nets	1
Objective Value	0.007445571
Mean Squared Error for Validation	2.9201402562

Iteration History

Iteration Number	Objective Function	Norm of Gradient	Loss	Validate Error	Step Size	Norm L1	Norm L2	Norm Maximum	Fit Error
1	0.055575	0.176714	0.054811	16.79691	0	15.93753	0.763306	0.066667	14.13496
2	0.040283	0.004111	0.039522	15.18386	0.979453	15.90880	0.761843	0.066951	10.19196
3	0.039979	0.004990	0.039235	15.06992	21	15.46631	0.743607	0.068300	10.11814
4	0.012115	0.064623	0.008979	4.631559	8.395674	45.93999	3.136635	0.856929	2.315444
5	0.012050	0.057190	0.008890	4.568114	0.002087	46.29671	3.160375	0.863465	2.292561
6	0.011720	0.036272	0.008624	4.153144	1	45.35487	3.095608	0.845005	2.224043
7	0.011420	0.017724	0.008439	4.077412	1	43.69028	2.980638	0.811340	2.176394
8	0.010852	0.044709	0.008557	4.218472	1	34.27831	2.294557	0.609633	2.206705
9	0.010438	0.019566	0.008069	3.870357	1	35.29760	2.368802	0.630178	2.080811
10	0.009799	0.012030	0.007740	3.624999	1	31.33528	2.058958	0.533070	1.995980
11	0.009407	0.017500	0.007601	3.584056	0.471163	28.23288	1.806046	0.448531	1.960211
12	0.008691	0.008926	0.007143	3.237078	1	25.50180	1.547827	0.417925	1.842174
13	0.008595	0.011554	0.007105	3.273865	0.333708	24.77158	1.489458	0.408422	1.832361
14	0.008259	0.022004	0.006783	3.090086	1	24.51895	1.475966	0.410441	1.749163
15	0.008091	0.008378	0.006677	3.098389	0.394350	23.73946	1.413858	0.401656	1.721951
16	0.008003	0.006333	0.006602	3.048834	1	23.60343	1.400885	0.403446	1.702539
17	0.007894	0.006600	0.006517	3.042751	1	23.33452	1.376346	0.403519	1.680677
18	0.007775	0.021423	0.006363	3.187524	1	23.92279	1.411855	0.409962	1.640963
19	0.007552	0.005109	0.006149	3.013543	1	23.75586	1.403266	0.412350	1.585646
20	0.007542	0.003681	0.006143	3.038754	0.287731	23.71000	1.399675	0.411051	1.584066
21	0.007539	0.001343	0.006139	3.033251	0.449050	23.70846	1.400116	0.411609	1.583197
22	0.007538	0.001529	0.006138	3.038583	1	23.71205	1.400455	0.411558	1.582791
23	0.007530	0.002299	0.006129	3.041075	1	23.68324	1.400467	0.411557	1.580643
24	0.007506	0.003898	0.006103	3.030961	1	23.58610	1.402949	0.413022	1.573784
25	0.007478	0.004160	0.006074	2.995946	1	23.44517	1.404534	0.415300	1.566345
26	0.007475	0.003118	0.006071	3.002382	0.075645	23.42455	1.404294	0.415440	1.565562
27	0.007453	0.002345	0.006051	2.948176	1	23.28916	1.402249	0.417514	1.560492
28	0.007452	0.001942	0.006050	2.944997	0.110261	23.27597	1.402060	0.417571	1.560144
29	0.007446	0.001502	0.006046	2.920140	1	23.20020	1.399424	0.418223	1.559201
30	0.007432	0.002195	0.006043	2.924769	1	22.93835	1.388999	0.418067	1.558341
31	0.007420	0.001290	0.006038	2.937230	1	22.81177	1.381801	0.417053	1.557084
32	0.007410	0.001647	0.006038	2.966920	1	22.64662	1.372487	0.415473	1.556986
33	0.007395	0.001516	0.006032	3.007425	1	22.42932	1.362949	0.414449	1.555560
34	0.007386	0.002567	0.006030	3.054353	0.390473	22.28657	1.356232	0.413991	1.554995

The optimization exited on validation criteria

Predicted Target Variable
Variable
P_next_year_log_export_value

Score Information for Training	
Number of Observations Read	1372
Number of Observations Used	1372
Mean Squared Error	1.5592010169

Score Information for Validation	
Number of Observations Read	392
Number of Observations Used	392
Mean Squared Error	2.9201402562

Fit Statistics

Target Name	Data Role	Partition Indicator	Formatted Partition
next_year_log_export_value	TRAIN	0	0
next_year_log_export_value	VALIDATE	1	1

Number of Observations	Average Squared Error	Divisor for ASE	Root Average Squared Error
1,372	1.5592	1,372	1.2487
392	2.9201	392	1.7088

Mean Absolute Error	Root Mean Absolute Error	Mean Squared Logarithmic Error	Root Mean Squared Logarithmic Error
0.7835	0.8852	0.0317	0.1781
1.0256	1.0127	0.1141	0.3378

R-Square
0.8469
0.8030

Network Diagram: Top 200 Weights

